

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Industrial and Systems Engineering

2. Subject Name: Optimization and Heuristics Methods L-T-P:3-0-0
Credits: 3

3. Pre-requisites: Operations Research II (IM21204)

4. Syllabus and reference books:

Syllabus: Classification of optimization problems. A brief review of conventional search algorithms Introduction to OHM. Simulated Annealing, Tabu Search, Genetic Algorithm, Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Artificial Bees Colony (ABC), Greedy Heuristics, Multi-Objective Optimization, NSGA approach for Multi-Objective Optimization. Fuzzy Logic- Fuzzy Sets and Relation, Fuzzification and Defuzzification , Membership Functions, Fuzzy Inferences

Space for entering the syllabus

Reference Books:

- 1) Handbook of Metaheuristics – Michel Gendreau and Jean-Yves Potvin
- 2) Multi-Objective Optimization using Evolutionary Algorithms – Kalyanmoy Deb
- 3) Fuzzy logic with Engineering applications - T J Ross

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Classification of optimization problems. A brief review of conventional search algorithms Introduction to OHM	2
2.	Simulated Annealing	3
3.	Tabu Search	3
4.	Genetic Algorithm	6
5.	Particle Swarm Optimization	3
6.	Ant Colony Optimization	3
7.	Artificial Bees Colony Optimization	3
8.	Multi Objective Optimization and NSGA	4
9.	Fuzzy Logic Introduction, Fuzzy Sets and Relation	4
10.	Fuzzification-Defuzzification and Membership Functions	5

11.	Fuzzy Inferences	3
Total number of hours		39